

Advanced (Habanero)

- Q1.** Solve the inequality $|x| < 3$.
 (A) $(-\infty, -3) \cup (3, \infty)$
 (B) $(-3, 3)$
 (C) $(-\infty, -3] \cup [3, \infty)$
 (D) $(-3, 0) \cup (0, 3)$
 (E) $[0, 3)$
- Q2.** Solve the inequality $|2x - 1| \geq 5$.
 (A) $(-\infty, -2] \cup [3, \infty)$
 (B) $(-2, 3)$
 (C) $(-\infty, -2) \cup (3, \infty)$
 (D) $[-2, 3]$
 (E) $(-\infty, 3] \cup [-2, \infty)$
- Q3.** Graph the solution set of $|x| > 2$ on a number line.
 (A) $(-\infty, -2) \cup (2, \infty)$
 (B) $(-2, 2)$
 (C) $(-\infty, -2] \cup [2, \infty)$
 (D) $[-2, 2]$
 (E) $(-2, 2)$
- Q4.** The temperature in a city is never more than 25°C above or below 0°C . Write an inequality to model this situation.
 (A) $|T| \leq 25$
 (B) $|T| \geq 25$
 (C) $T \leq 25$
 (D) $T \geq 25$
 (E) $T = 25$
- Q5.** Solve the inequality $|x - 2| < 4$.
 (A) $(-2, 6)$
 (B) $(-\infty, -2) \cup (6, \infty)$
 (C) $[-2, 6]$
 (D) $(-2, 2) \cup (2, 6)$
 (E) $(-\infty, -2] \cup [6, \infty)$
- Q6.** The cost of a taxi ride is never more than 15 above or below 10. Write an inequality to model this situation.
 (A) $|C - 10| \leq 15$
 (B) $|C - 10| \geq 15$
 (C) $C \leq 15$
 (D) $C \geq 15$
 (E) $C = 15$
- Q7.** Solve the inequality $|3x + 2| > 7$.
 (A) $(-\infty, -3) \cup (1.6667, \infty)$
 (B) $(-3, 1.6667)$
 (C) $(-\infty, -3] \cup [1.6667, \infty)$
 (D) $(-3, 0) \cup (0, 1.6667)$
 (E) $[0, 1.6667)$
- Q8.** Graph the solution set of $|x + 1| \leq 2$ on a number line.
 (A) $[-3, 1]$
 (B) $(-\infty, -3) \cup (1, \infty)$
 (C) $(-3, 1)$
 (D) $[-3, 1)$
 (E) $(-3, 1]$

Open Ended Questions (FRQ)

- Q9.** A company packages coffee in bags that hold between 10 and 14 ounces of coffee. Write an inequality to model the amount of coffee x in each bag and solve for x . Show your work.
- Q10.** The length of a shadow of a flagpole is never more than 5 meters longer or shorter than the height of the flagpole. If the length of the shadow is x meters and the height of the flagpole is 10 meters, write an inequality to model this situation and solve for x . Show your work.

Chef's Tips

- Tip 1: Make sure to read each question carefully.
- Tip 2: Use a number line to visualize the solution sets.
- Tip 3: Check your work by plugging in values from your solution set.